

Chat Stuff

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1 summary april 19

Title (TBD) Mass as a Sink of Zero-Point Radiation: A Classical Alternative to General Relativity

Abstract (One-paragraph summary of the motivation, hypothesis, method, and main results.)

1. Introduction Historical background: failures of GR/QM unification

Motivation for a new approach

Preview of assumptions and structure of paper

Summary of results: reproducing classical GR tests, recovering Dirac

2. Assumptions and Hypothesis Radiation field assumed to be filled with stochastic background (ZPF)

Mass sinks radiation $\Rightarrow$ induces directional momentum flow

Key hypothesis:

Connection to zero-point energy and cutoff frequency

3. Maxwell-Style Equations for Gravity Modified field equations (Heaviside/Boyer-style)

Decomposition into irrotational and solenoidal parts

Field solutions, energy/momentum densities

Setup for duality and spinor formulation later

4. Stochastic Electrodynamics Zero-point field background

Historical context (Boyer, Casimir, Puthoff)

Main equations and assumptions

Why ZPF is relevant to gravitational interaction

5. Mass and Radiation: A Physical Model 5.1 Radiation Sink Intuition

Imagine a warm iron ball cooling toward absolute zero

Postulate: cold mass still interacts with background ZPF

Visual and physical motivation for the sink hypothesis

5.2 Dual Cutoffs: Radius and Frequency Lower bound: minimum radius via gravitational self-energy

Upper bound: maximum absorbable frequency from energy balance

Leads to derivation of directional momentum flow

6. Field Energy and Momentum Balance Field energy provides stability against gravitational collapse

Radiation absorption balances self-attraction

Gravitational analog of Ohm's Law

Momentum conservation yields emergent conductivity

7. Classical Tests of Gravity 7.1 Perihelion Precession of Mercury Gravito-magnetic correction

Effective potential

Agreement with GR

7.2 Light Bending Around a Mass Momentum transfer to photons

Path integral treatment

Reproduces GR deflection angle

7.3 Gravitational Redshift Momentum gain of falling photons

Matches GR blue-shift at lowest order

8. Duality and the Dirac Equation	8.1 Dual Transform (Jackson)	Electro-
		magnetic symmetry
	Rotation between E/B components	
8.2 Spinor Representation	Gamma matrix formulation	
	Spinor structure from field tensors	
8.3 Emergence of Dirac Field equations in spinor form		
	Recovery of Dirac equation from classical gravity	
9. Discussion and Outlook	Interpretation: Is this a force? A flow?	
	Experimental testability (e.g. high-frequency deviations?)	
	Limitations: no entanglement, classical field only	
	Future directions: nonlinear extensions, entropic gravity, quantization?	
Appendices (Optional)	A. Detailed derivations	
	B. Glossary of symbols	
	C. References to SED literature	

2 Title April 18

2.1 Formal and Academic

A Classical Field Theory of Gravity with Radiation Momentum Transfer
Mass as a Sink of Zero-Point Radiation: A Classical Alternative to General
Relativity
From Maxwell to Gravity: A Classical Radiation Model of Mass
Gravitational Phenomena from Radiation Momentum Flow

2.2 Conceptual / Thematic

The Radiation That Binds
Gravity Without Curvature
Mass Absorbs Light
A Field of Sinks
The Weight of Light
Punchy and Bold What If Mass Eats the Vacuum?
Gravity, Rebooted
No Curves, No Quanta
The Maxwellian Gravity Hypothesis
Not Bent—Absorbed

2.3 Philosophical / Ambitious

Toward a Classical Foundation of Gravity and Mass
Can Gravity Be Maxwellian?
Sinking the Quantum Vacuum
Reconstructing Gravity from Field Flow
A New Story of Mass

3 Introduction April 19

4 Introduction

The 20th century gave us two triumphs of theoretical physics: General Relativity (GR) and Quantum Mechanics (QM). GR describes the curvature of spacetime caused by mass and energy; QM governs the probabilistic behavior of particles and fields at the smallest scales. Each is spectacularly successful within its domain. But attempts to unify them—into quantum gravity, a theory of everything, or even just a consistent description of black holes—have proven stubbornly elusive.

In particular, gravity resists quantization. Standard field-theoretic approaches suffer from non-renormalizability, and more radical solutions like string theory or loop quantum gravity have not yielded unique, testable predictions. Meanwhile, GR struggles with singularities and fails to account for phenomena like the cosmological constant problem. These tensions suggest that the current frameworks may be incomplete—or even misframed.

This paper proposes a classical alternative to gravity, inspired by the structure of Maxwell's equations and informed by insights from stochastic electrodynamics (SED). The core hypothesis is that **mass sinks zero-point radiation**, creating a directional momentum flux field analogous to an electric field from a charged particle. Rather than curving spacetime, gravity in this framework emerges from a field-mediated interaction between matter and a stochastic radiation background.

We construct this theory from first principles using modified Maxwell-style equations, introduce a cutoff frequency tied to the mass of the particle, and develop a consistent model of massive particles as absorbers of radiation. From this starting point, we derive three classical tests of GR (perihelion precession, light bending, and gravitational redshift) and find that our results match those of GR at lowest order. Finally, we show that Maxwell's equations in this framework can be re-written as a spinor equation, yielding the Dirac equation as an emergent structure.

4.1 A Physical Picture: The Cold Iron Ball

Imagine holding a small iron ball, still warm from the oven, glowing faintly red as it radiates heat. Now imagine placing it in the vacuum of deep space. As it cools, the infrared radiation streaming off its surface begins to dwindle. The ball gets colder, eventually reaching absolute zero.

But now imagine that even at zero Kelvin, the universe is not empty. It is filled with a faint hiss of zero-point radiation—a stochastic electromagnetic background that never goes away. The iron ball no longer glows, but it still absorbs. It acts like a drain, a sink, pulling in tiny fragments of momentum from the vacuum field. The surrounding radiation field adjusts in response.

In this paper, we propose that **this flow of momentum—always inward, always present—is gravity.** Mass interacts with the zero-point field not by

bending spacetime, but by distorting the momentum landscape around it. The radiation field, shaped by the presence of mass, guides other particles and light beams as they pass nearby. The result is indistinguishable, at least at low energy, from Einstein’s gravity. But the mechanism is entirely classical.

The math begins now.

— and a different versions

5 Introduction

The search for a unified theory of quantum mechanics and gravity remains one of the most elusive goals in theoretical physics. Over a century after Einstein’s general relativity and the formalization of quantum field theory, no consistent framework has emerged that can fully reconcile the two. Quantum gravity proposals have flourished — from string theory to loop quantum gravity — yet none have produced unambiguous experimental validation. Meanwhile, general relativity and quantum electrodynamics each function remarkably well in their respective domains, leaving theorists caught between two incompatible successes.

This paper proposes a radically different approach. Rather than quantizing gravity or geometrizing quantum mechanics, we hypothesize a classical field-theoretic mechanism by which gravitational behavior emerges from radiation momentum transfer. The core idea is that mass acts as a *sink* for zero-point radiation, inducing a directional flow of momentum in the surrounding field. This momentum flow behaves, to leading order, like Newtonian gravity — and at higher order, replicates the classical tests of general relativity. Most remarkably, by expressing the theory in spinor form, the equations reduce to the Dirac equation for a massive particle.

The theory is formulated in terms of modified Maxwell-like equations, inspired by Heaviside’s gravitoelectromagnetism and enriched by the stochastic background field ideas of Boyer and the stochastic electrodynamics (SED) community. Throughout the paper, we maintain a classical framework: no quantization, no operator formalism, no probabilistic interpretation — yet key features of quantum and gravitational behavior emerge as natural consequences of the dynamics.

5.1 A Physical Picture: The Cold Iron Ball

Before launching into equations, we offer an image. Imagine a warm iron ball removed from a furnace, glowing red as it radiates heat. Now imagine the same ball, perfectly cold — at absolute zero — placed in a universe filled with fluctuating electromagnetic radiation. In that frigid silence, the ball doesn’t glow — it absorbs.

Our proposal is this: even at zero temperature, the ball — by virtue of its mass — continues to interact with the ambient field. It sinks energy. And this

absorption is not passive. It draws in momentum from the field, creating a radial flow toward the mass.

This is the metaphorical heart of the theory. Just as electric charge sources field lines, and magnetic poles source curls, we posit that mass sources a sink in the momentum field of zero-point radiation. The rest of the paper builds on this intuition.

6 Assumptions and Hypothesis

The theory rests on two central assumptions and one key hypothesis.

6.1 Assumption 1: The Vacuum is Not Empty

We assume that empty space is filled with a stochastic electromagnetic background — a zero-point field (ZPF) — with spectral energy density:

$$\rho(\omega) = \frac{1}{2}\hbar\omega$$

This background field is Lorentz-invariant, homogeneous, and isotropic, as developed in the framework of stochastic electrodynamics. It corresponds to a classical analogue of the quantum vacuum, but we treat it as a physically real radiation field with classical dynamics.

6.2 Assumption 2: Classical Field Theory is Valid

We adopt classical field theory as the mathematical framework — not quantum field theory, and not curved spacetime. Specifically, we assume that the structure of electromagnetic field theory (Maxwell’s equations) provides the correct blueprint for gravitational dynamics, albeit with certain modifications. Fields propagate in flat spacetime and interact linearly, unless otherwise specified.

6.3 Hypothesis Mass Absorbs Radiation

As suggested in the introduction, we hypothesize that massive particles are not transparent to the ZPF. Instead, they absorb radiation — not randomly, but in a directional fashion, proportional to the local field. This absorption gives rise to a net momentum flux in the radiation field.

The core hypothesis is that this directional absorption leads to a net inward flow of momentum, governed by:

$$\mathbf{p}(r) = -\hat{\mathbf{r}} \frac{Gm\hbar}{r^2}$$

This is not a force law — it is a momentum flux density induced by the interaction between the mass and the background field. Unlike Newton’s law of

gravitation, this expression is not derived from an action-at-a-distance potential but from a field-theoretic interaction with the vacuum.

We interpret this as the gravitational effect of the mass: it does not pull on other bodies directly, but modifies the momentum structure of the radiation field around it. Other particles traveling through this field experience deviations consistent with classical gravity.

The remainder of the paper explores the consequences of this hypothesis. We derive the necessary modifications to Maxwell's equations, define a consistent model of a massive particle, and demonstrate how this mechanism can replicate the classical tests of general relativity — all while remaining within a classical framework.

7 radiation sink intuition april 19

Radiation Sink Intuition To motivate our physical model, consider a familiar example: a warm iron ball, freshly removed from an oven. It radiates heat in the form of infrared photons, gradually cooling as energy escapes into its environment. As it approaches absolute zero, the thermal radiation diminishes—but not entirely. According to both classical and quantum considerations, the universe is never truly empty: the zero-point field (ZPF) remains, even in a vacuum at zero temperature.

Our core hypothesis is that mass continues to interact with this omnipresent radiation field, even when all conventional thermal energy is gone. The cold iron ball doesn't just sit inertly in the vacuum—it absorbs from the ZPF.

This mental picture provides an anchor for the rest of the theory. If mass always interacts with the ZPF—quietly, continuously—then this interaction must manifest physically. Our proposal is that this manifests as an inward-pointing momentum density, scaling as an inverse square law. That is:

This is not derived yet—it is proposed. But it invites us to ask: can such a flow be consistent with known energy and momentum conservation laws? Can it stabilize a massive object against collapse? Can it explain gravitational phenomena?

In the following section, we begin to quantify this picture by examining what sets the limits: How small can the mass shrink before gravitational collapse dominates? And up to what frequency does the mass interact with the ZPF?

8 massive particle april 18

9 Mass as a Radiation Sink: Core Hypothesis

9.1 Directional Momentum Flow

We now introduce the central hypothesis of this theory: **mass acts as a sink for zero-point radiation**, producing a directional inflow of momentum from the surrounding field. This effect mimics a gravitational force and emerges naturally from the structure of the field equations.

From the previous section, we obtained the general form of the momentum in the radiation field:

$$\mathbf{P}_{\text{field}} = \frac{1}{8\pi} \int_V \mathbf{E} \times \mathbf{B}^* d^3x = -\hat{\mathbf{r}} \sum_{l,m} \int_V \frac{1}{4\pi} \frac{|a_E|^2 |\mathbf{X}_{lm}|^2}{k^2 r^2} d^3x$$

Absorbing the constants, we write this compactly as:

$$\mathbf{p} = -\hat{\mathbf{r}} \frac{\alpha}{r^2}$$

We will derive the constant α in the next section. When this is done, we arrive at the key postulate of this paper:

$$\boxed{\mathbf{p} = -\hat{\mathbf{r}} \frac{Gm\hbar}{r^2}}$$

This expression forms the foundation of our model: *mass induces a momentum flow in the surrounding zero-point field*, with a structure formally analogous to Newtonian gravity.

9.2 Cutoff Frequency

A further assumption is required: that this radiation sink effect only holds up to a maximum frequency. Physically, this can be interpreted as a natural limit to the energy that a given mass can absorb.

Using a simple energy balance:

$$U = m = \hbar\omega_{\text{max}} \quad \Rightarrow \quad \omega_{\text{max}} = \frac{m}{\hbar}$$

This cutoff ensures convergence of integrals involving radiation absorption. For ordinary matter, the corresponding frequency is many orders of magnitude above familiar scales.

9.3 Model of a Massive Particle

To proceed, we define what we mean by a *massive particle*. The most direct classical model is a spherically symmetric mass distribution. While not a literal description of elementary particles, it provides a tractable mathematical structure.

Following Rohrlich, we consider a spherical distribution of mass density $\rho(\mathbf{x})$, with total mass:

$$m = \int_V \rho(\mathbf{x}) d^3x$$

In the absence of a counterbalancing force, the mass distribution would gravitationally collapse to a point. This classical result motivates the need for an external stabilizing mechanism—in our case, radiation absorption.

The gravitational self-energy of a spherically symmetric distribution is:

$$U = -\frac{1}{2} \int \frac{Gm\rho(\mathbf{x})}{r} d^3x$$

Taking the simplest case—a thin spherical shell of radius R —yields:

$$m = \frac{Gm^2}{2R} \quad \Rightarrow \quad R = \frac{Gm}{2}$$

Other distributions yield similar scaling. We adopt the compact form:

$$R \sim Gm$$

This defines the minimal physical radius of a classical massive particle in this framework.

Comparison to Physical Systems

We define the dimensionless compactness parameter:

$$\Phi = \frac{Gm}{Rc^2}$$

and compare theoretical and physical systems:

Mathematical limit:	$\Phi = 1$
Black hole:	$\Phi = 1$
Sun:	$\Phi \approx 2 \times 10^{-6}$
Earth:	$\Phi \approx 7 \times 10^{-10}$
Proton:	$\Phi \approx 1 \times 10^{-38}$

Real objects are far from the $\Phi = 1$ regime—except black holes. This scaling forms the basis for exploring high-density limits in our theory.

9.4 Energy Conservation and Stability

We now demonstrate that a massive particle can remain stable, due to energy exchange with the radiation field.

Field Energy

The total energy stored in the surrounding radiation field is:

$$U = \frac{1}{16\pi} \int (|\mathbf{E}|^2 + |\mathbf{B}|^2) d^3x$$

From our earlier solutions, this becomes:

$$U = \sum_{l,m} \int_V \frac{1}{4\pi} \frac{|a_E|^2 |\mathbf{X}_{lm}|^2}{k^2 r^2} d^3x$$

Defining the energy density:

$$u(r) = \sum_{l,m} \frac{1}{4\pi} \frac{|a_E|^2 |\mathbf{X}_{lm}|^2}{k^2 r^2} = \frac{\alpha}{r^2}$$

and applying the frequency cutoff:

$$u(r) = \int_0^{m/\hbar} \frac{\alpha}{r^2} d\omega = \frac{\alpha m}{\hbar r^2}$$

Total Energy and Balance

The total energy is then:

$$U = \int_V u(r) d^3x = -\frac{Gm^2}{R} + \frac{4\pi\alpha m R}{\hbar}$$

Enforcing energy conservation:

$$U_{\text{total}} = 0 \quad \Rightarrow \quad \alpha = \frac{1}{4\pi R^2} Gm\hbar$$

Substitute into $u(r)$:

$$u(r) = \frac{Gm\hbar}{4\pi R^2 r^2}$$

This supports the view that the field dynamically stabilizes the mass by opposing gravitational collapse.

9.5 Momentum Conservation and Effective Conductivity

To complete the consistency check, we examine momentum conservation and derive an effective Ohm's law.

Momentum Balance

From Jackson Eq. 6.114:

$$\int_V (\rho \mathbf{E}_I + \mathbf{J}_S \times \mathbf{B}_S) d^3x = 0$$

Ohm's Law

Assume:

$$\mathbf{J}_S = \sigma \mathbf{E}_S$$

and harmonic time dependence. Evaluate:

$$-\hat{\mathbf{r}} \int_V \left(m \delta(\mathbf{x}) \cdot \frac{Gm}{r^2} - \sigma \cdot \frac{Gm\hbar}{r^2} \right) d^3x = 0$$

Solving:

$$\sigma = \frac{m}{\hbar} \delta(\mathbf{x}) \quad \text{or} \quad \sigma = \frac{m}{4\pi\hbar} \delta(\mathbf{x})$$

Interpretation

This implies the massive particle has a localized delta-function conductivity at the origin. This emerges naturally from conservation laws and further supports the central hypothesis: *mass interacts with the radiation field as a sink, leading to a self-consistent stabilization of the particle.*

10 Two cutoffs april 19

10.1 Cutoffs

A central feature of this theory is the existence of two natural cutoffs: one in *space* and one in *frequency*. These boundaries are not imposed by hand but arise from physical consistency: the spatial cutoff prevents gravitational collapse, while the frequency cutoff ensures finite absorption from the zero-point field.

Lower Bound: Minimum Radius via Gravitational Self-Energy

From classical considerations, any spherically symmetric mass distribution has a gravitational self-energy:

$$U = -\frac{1}{2} \int \frac{Gm\rho(\mathbf{x})}{r} d^3x$$

For a spherical shell of radius R , this leads to:

$$R = \frac{Gm}{2}$$

Other distributions differ only by numerical prefactors. In general, we find that a stable radius must satisfy:

$$RGm$$

We interpret this as a **minimum radius** for any particle with mass m . Unlike the Schwarzschild radius (which assumes collapse), this lower bound emerges as a stability criterion in the presence of zero-point radiation pressure.

Upper Bound: Cutoff Frequency via Energy Matching

On the other hand, a mass cannot absorb arbitrarily high-frequency radiation. The frequency cutoff is set by energy conservation:

$$U = m = \hbar\omega_{\max} \quad \Rightarrow \quad \omega_{\max} = \frac{m}{\hbar}$$

This cutoff controls the integration limits in our field energy and momentum densities. While this frequency is extremely high for typical masses, it plays a crucial theoretical role: it prevents divergences and ensures the total absorbed radiation remains finite.

Together, these two cutoffs—one in space, one in frequency—define the domain over which the mass interacts with the field. They also point toward a deeper structure: a kind of “resolution scale” for gravity, where classical continuous absorption breaks down and new physics (perhaps quantum or nonlinear) may begin to emerge.

$$r = \frac{Gm}{k} = \frac{m}{h} \quad m = \frac{r}{G} \quad k = \frac{r}{Gh}$$

11 energy and mom april 19

12 Field Energy and Momentum Balance

12.1 Stability from Field Energy

We now evaluate whether the inward radiation flow, hypothesized in Section 2, can provide a stabilizing mechanism for a massive particle. Specifically, we ask: can the absorbed energy from the zero-point field counteract the gravitational self-energy that would otherwise lead to collapse?

The total field energy is given by:

$$U = \frac{1}{16\pi} \int (|\mathbf{E}|^2 + |\mathbf{B}|^2) d^3x$$

From earlier results, the energy density near a massive particle absorbing zero-point radiation falls off as $1/r^2$. Including the cutoff frequency $\omega_{\max} = m/\hbar$, we have:

$$u(r) = \int_0^{m/\hbar} \frac{\alpha}{r^2} d\omega = \frac{\alpha m}{\hbar r^2}$$

Integrating over all space, assuming spherical symmetry:

$$U_{\text{rad}} = \int_0^\infty u(r) \cdot 4\pi r^2 dr = \int_0^\infty \frac{4\pi\alpha m}{\hbar} dr$$

This diverges unless we impose a lower bound, namely the minimum radius of the particle $R = Gm$. Truncating the integral from R to ∞ , we obtain:

$$U_{\text{rad}} = \frac{4\pi\alpha m}{\hbar} \int_R^\infty dr = \infty$$

To regularize this, we instead take the volume integral up to some finite domain or assume that most of the energy is absorbed near R . The key is not the total energy, but the ****balance**** between the field energy and the self-gravitational energy of the particle:

$$U_{\text{grav}} = -\frac{Gm^2}{R}$$

$$U_{\text{rad}} = \frac{4\pi\alpha m R}{\hbar}$$

Balancing these energies (assuming $U_{\text{total}} = 0$) gives:

$$-\frac{Gm^2}{R} + \frac{4\pi\alpha m R}{\hbar} = 0 \quad \Rightarrow \quad \alpha = \frac{Gm\hbar}{4\pi R^2}$$

Substituting back, we find the energy density is:

$$u(r) = \frac{Gm\hbar}{4\pi R^2 r^2}$$

This shows that the particle's own radiation environment acts to resist collapse, provided the momentum flow structure is accepted. The directionality of $\mathbf{p} \propto -\hat{\mathbf{r}}$ also implies an inward pressure that balances gravitational pull.

12.2 Momentum Conservation and “Ohm’s Law”

We now turn to momentum conservation. The total momentum exchange in the system should obey:

$$\int_V (\rho \mathbf{E}_I + \mathbf{J}_S \times \mathbf{B}_S) d^3x = 0$$

This relation balances the impulse from the irrotational field (associated with gravitational mass) against the magnetic impulse from the solenoidal radiation field.

Assuming the solenoidal current obeys a linear response:

$$\mathbf{J}_S = \sigma \mathbf{E}_S$$

and harmonic time dependence $e^{-i\omega t}$, the field momentum becomes:

$$-\hat{\mathbf{r}} \int \left(m\delta(\mathbf{x}) \cdot \frac{Gm}{r^2} - \sigma \cdot \frac{Gm\hbar}{r^2} \right) d^3x = 0$$

Solving gives:

$$\sigma = \frac{m}{\hbar} \delta(\mathbf{x})$$

This expression defines a delta-function conductivity concentrated at the origin — an effective Ohm’s law in which the mass acts as a point-like, perfect absorber for incoming radiation. Including conventional normalization gives:

$$\sigma = \frac{m}{4\pi\hbar} \delta(\mathbf{x})$$

This result is not imposed, but emerges from requiring consistency with the momentum flux model.

12.3 Summary

Together, the field energy and momentum conservation arguments demonstrate that the hypothesized momentum inflow can act as a self-consistent stabilizing structure. A massive particle that sinks zero-point radiation does not collapse — not due to internal structure, but due to interaction with the vacuum field itself.

13 conclusion april 18

14 Discussion and Outlook

14.1 Is This a Force?

At first glance, the directional momentum flow described in this theory might resemble a gravitational force. But it is not. The core hypothesis postulates a field of momentum density sourced by mass and arising from asymmetric absorption of zero-point radiation. The resulting momentum transfer acts on light but cannot be described by $\mathbf{F} = m\mathbf{a}$, nor does it require mass to accelerate—it is a field interaction, not a mechanical push.

This distinction is important. In this model, light does not bend due to spacetime curvature or inertial trajectories, but because its momentum is shifted by a background field. Whether this qualifies as a “force” depends on one’s metaphysical commitments; mathematically, it is simply a field-driven momentum evolution.

14.2 Experimental Implications and Testability

At low frequencies, this theory reproduces the standard predictions of general relativity to first order: the perihelion precession of Mercury, bending of light, and gravitational redshift. However, the model contains a natural high-frequency cutoff, $\omega_{\text{max}} = m/\hbar$, that suggests potential deviations from general relativity in high-energy regimes.

This may provide a testable prediction: does light near extremely dense masses show slight deviation from GR’s trajectory at sub-Planckian wavelengths? Could precise interferometry or black hole lensing data reveal anomalies that hint at a momentum field with a frequency cutoff?

These are open questions, but the existence of a scale—where the theory must break or transition—is, arguably, a feature not a bug.

14.3 Limitations of the Current Framework

This theory is explicitly classical. It does not attempt to reproduce or explain entanglement, Bell violations, or other nonlocal quantum effects. It treats zero-point radiation as a real field, but it does not incorporate quantization of the field itself.

Likewise, no mechanism for spin, gauge symmetry, or quantized interaction vertices is included. The model focuses narrowly on reproducing gravitational behavior via field theory, not on building a replacement for quantum mechanics or QFT. The theory operates best in the semi-classical, single-particle domain.

14.4 Future Directions

A number of extensions suggest themselves. First, the behavior of nonlinear field interactions has not yet been analyzed—how do multiple nearby masses interact? Do waves interfere constructively or destructively? Is the radiation back-reaction linear?

Second, it would be valuable to explore whether a thermodynamic or entropic gravity formulation could emerge from this momentum flux model. Given that the field acts like a flow of energy and momentum, some form of entropic argument may provide further grounding.

Finally, full quantization is an open frontier. Is there a path from this classical field theory to a quantum field theory of gravity that retains the core insight—that mass sinks radiation—and reconciles it with quantum field dynamics?

These questions remain unanswered, but the framework introduced here offers a fresh angle and a stable platform from which to explore them.